QUESTION ANSWERING USING OLLAMA AND PYTHON

Question 6 – Implementation

1. Question

Demonstrate question answering using a local Large Language Model through Ollama and Python.

2. Aim

To demonstrate a Python-based question-answering system that sends user questions to a locally running Llama 3.2 Large Language Model through Ollama and displays the generated answers in the terminal.

3. Technologies Used

- Python 3.12.10
- Ollama 0.33.2
- Llama 3.2 – local Large Language Model
- Ollama Python library
- VS Code

4. Project Structure

```
Local_LLM_Text_Generation/
├── venv/ – Python virtual environment
├── app.py – Question 1 text generation
├── summarization.py – Question 2 text summarization
├── qa.py – Question 3 Streamlit question answering
├── translation_paraphrasing.py – Question 4 translation and paraphrasing
├── ollama_text_generation.py – Question 5 Ollama and Python text generation
└── ollama_qa.py – Question 6 Ollama and Python question answering
```

5. Implementation Steps

1. Use the existing Local_LLM_Text_Generation project and virtual environment.
2. Ensure Ollama is installed and the Llama 3.2 model is available locally.
3. Import the ollama Python library.
4. Create an ask_question() function to communicate with the local Llama 3.2 model.
5. Accept a question from the user using the Python input() function.
6. Place the question inside a structured question-answering prompt.
7. Send the prompt to Ollama using ollama.chat().
8. Receive the generated answer from the local model.
9. Display the answer in the Python terminal.
10. Run the program using python ollama_qa.py.

6. Python Implementation

```
import ollama


def ask_question(question):
    response = ollama.chat(
        model="llama3.2",
        messages=[
            {
                "role": "user",
                "content": f"""
You are a helpful question-answering assistant.

Answer the following question clearly and accurately.
Use simple language and provide enough explanation.

Question:
{question}

Answer:
"""
            }
        ]
    )

    return response["message"]["content"]


print("=" * 60)
print("LOCAL LLM QUESTION ANSWERING USING OLLAMA")
print("=" * 60)

question = input("\nEnter your question: ")

if question.strip():

    print("\nGenerating answer…\n")

    answer = ask_question(question)

    print("Answer:")
    print("-" * 60)
    print(answer)
    print("-" * 60)

else:
    print("\nPlease enter a valid question.")
```

7. Working Principle

The program accepts a natural-language question from the user and constructs a question-answering prompt. The prompt is sent to Ollama, which runs the Llama 3.2 model locally. The generated answer is returned to the Python application and printed in the terminal.

8. System Workflow

11. User enters a question.
12. Python receives the question.
13. The question is inserted into the structured prompt.
14. Python sends the prompt to Ollama.
15. Ollama invokes the local Llama 3.2 model.
16. Llama 3.2 generates an answer.
17. The response is returned to Python.
18. Python displays the answer in the terminal.

9. Input

Test question: What is Artificial Intelligence?

10. Output

The local Llama 3.2 model generated an explanation of Artificial Intelligence. The response described AI as technology that enables computers and machines to perform human-like tasks, explained the role of algorithms and data, and provided examples such as virtual assistants, image recognition, chatbots, and predictive analytics.

11. Demonstrated Terminal Output

```
============================================================
LOCAL LLM QUESTION ANSWERING USING OLLAMA
============================================================

Enter your question: What is Artificial Intelligence?

Generating answer…

Answer:
------------------------------------------------------------
Artificial Intelligence (AI) is a type of technology that
helps computers and machines think and learn like humans do.

The response explained how AI uses algorithms and data to
make decisions and improve its performance. It also gave
examples including virtual assistants, image recognition,
chatbots, and predictive analytics.
------------------------------------------------------------
```

12. Result

Thus, question answering using a local Large Language Model was successfully demonstrated through Ollama and Python. The Llama 3.2 model processed the user question locally and generated a meaningful answer that was displayed in the terminal.

13. Conclusion

This implementation demonstrates how Python can interact directly with a locally running Large Language Model through Ollama for question answering. It provides a simple local alternative for experimenting with LLM-based question-answering without requiring a cloud-based API.